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CODETANTRA

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1.1.1. Calculate Momentum03:42

Write a program that accepts the mass of an object (in kilograms) and its velocity (in meters per second), then calculates and displays the momentum of the object. The momentum p is calculated using the formula:

$$p = m \times v$$

where:
 m is the mass of the object (in kilograms).
 v is the velocity of the object (in meters per second).

Input Format:

- A single floating-point number representing the mass of the object in kilograms.
- A single floating-point number representing the velocity of the object in meters per second.

Output Format:

- The output will display calculated momentum with appropriate units (kgm/s) (rounded up to 2 decimal places).

Sample Test Cases

calculate...

```
1 #Type Content here...
2 mass = float(input())
3 velocity = float(input())
4 momentum = mass * velocity
5 print(f"momentum: .2f)kgm/s")
```

TerminalTest cases

< PREV

RESET

SUBMIT

NEXT >

1.1.2. Conditional Calculation Based on the Number of Digits

13.06

Write a Python program that accepts an integer n as input. Depending on the number of digits in n .

Constraints: $1 \leq n \leq 999$ **Input Format:**

- The input consists of a single integer n .

Output Format:

- If n is a single-digit number, print its square.
- If n is a two-digit number, print its square root (rounded to two decimal places).
- If n is a three-digit number, print its cube root (rounded to two decimal places).
- Else print "Invalid".

Sample Test Cases

+

condition...

SUBMIT

```
1 n=int(input())
2 if n<10:
3     print(n*n)
4 elif 10<=n<=99:
5     a=n**0.5
6     print("{:.2f}".format(a))
7 elif 100<=n<=999:
8     b=n**(1/3)
9     print("{:.2f}".format(b))
10 else:
11     print("Invalid")
```

Terminal

Test cases

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1.1.3. Days between Two Dates08:12

Write a Python program to calculate number of days between two dates.

Input Format:

- The first line contains the first date in the format `YYYY-MM-DD`.
- The second line contains the second date in the format `YYYY-MM-DD`.

Output Format:

- An integer representing the number of days between the given dates.

Note:

- The first date should always be considered earlier than the second date.

Sample Test Cases

noOfDays...

```
1 from datetime import date
2 from datetime import datetime
3 date1 = input()
4 date2 = input()
5 d1 = datetime.strptime(date1, "%Y-%m-%d")
6 d2 = datetime.strptime(date2, "%Y-%m-%d")
7 difference = d2 - d1
8 print(difference.days)
9 #write your code here...
```

TerminalTest cases

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1.1.4. Reverse a Number

57.41

AA

Write a Python program to reverse the digits of the number and print the reversed number.

Input Format:

The input is a single integer.

Output Format:

The output prints a single integer, which is the reversed number.

Sample Test Cases

+

reverseN...

SUBMIT

1# Read input

2num = int(input())

3

4# Handle negative numbers

5sign = -1 if num < 0 else 1

6num = abs(num)

7

8# Reverse digits

9rev = 0

10while num > 0:

11 digit = num % 10

12 rev = rev * 10 + digit

13 num = num // 10

14

15# Apply sign back

16rev *= sign

17

18# Print result

19print(rev)

TerminalTest cases

< PREV

RESET

SUBMIT

NEXT >

1.1.5. Multiplication Table 03:18

Write a Python program that takes an integer as input and prints the multiplication table for that integer from 1 to 10.

Input Format:

- The first line of input contains an integer that represents the number for which the multiplication table is to be printed.

Output Format:

- Print the multiplication table for the given number in the format:

<number> x <num1> = <num2>

Note:

- The character "x" is used in the output to represent multiplication.
- Refer to the visible test-cases for more clarity.

Sample Test Cases +

multipl...

1 #write your code here...
2 n=int(input())
3 for i in range(1,11):
4 print(n,"x",i,"=",n*i)
5

Terminal Test cases

< PREV RESET SUBMIT NEXT >

1.2.1. Pass or Fail 26/17

Write a Python program that accepts the number of courses and the marks of a student in those courses.

The grade is determined based on the aggregate percentage:

- If the aggregate percentage is greater than 75, the grade is Distinction.
- If the aggregate percentage is greater than or equal to 60 but less than 75, the grade is First Division.
- If the aggregate percentage is greater than or equal to 50 but less than 60, the grade is Second Division.
- If the aggregate percentage is greater than or equal to 40 but less than 50, the grade is Third Division.

Input Format:

- The first input will be an integer n , the number of courses.
- The second input will be n integers representing the marks of the student in each of the n courses, separated by a space.

Output Format:

If the student passes all courses:

- Print the aggregate percentage (formatted to two decimal places).
- Print the grade based on the aggregate percentage.

If the student fails any course (marks < 40 in any course), print:

- "Fail".

Note:

- Aggregate Percentage: Average performance across all courses, calculated by summing all marks and dividing by the number of courses.

Sample Test Cases +

```
passorFa... SUBMIT
1 # write your code here
2 def calculate_student_grade():
3     try:
4         # Input: Number of courses
5         n = int(input())
6
7         # Input: Marks separated by space, converted to a list of integers
8         marks = list(map(int, input().split()))
9
10        # Safety check: Ensure we have the correct number of marks
11        if len(marks) != n:
12            return
13
14        # Check for failure: If any course mark is below 40
15        for m in marks:
16            if m < 40:
17                print("Fail")
18                return
19
20        # Calculate Aggregate Percentage (Average)
21        aggregate = sum(marks) / n
22
23        # Output the aggregate formatted to 2 decimal places
24        print(f"Aggregate Percentage: {aggregate:.2f}")
25
26        # Determine the Grade based on the aggregate percentage
27        if aggregate > 75:
28            print("Grade: Distinction")
29        elif aggregate >= 60:
```

1.2.2. Fibonacci Series

Write a Python program that uses recursion to print the first n terms of the Fibonacci series.

Input Format:

- A single integer n representing the number of terms to generate.

Output Format:

- A single line containing the first n terms of the Fibonacci sequence, separated by spaces.

Sample Test Cases

fibonacci...

```
1  '''def fibonacci(n):
2
3  --># write the code..
4
5  -->return fibonacci(-->)
6
7
8  n = int(input())
9  for i in range(1, n + 1):
10     print(fibonacci(i), end=" ")
11
12  def fibonacci(n):
13     # Base Case: return n if it is 0 or 1
14     if n <= 1:
15         return n
16     # Recursive Step
17     return fibonacci(n - 1) + fibonacci(n - 2)
18
19  def print_series():
20     try:
21         n_terms = int(input())
22
23         if n_terms <= 0:
24             return
25
26         for i in range(n_terms):
27             # Using end=" " ensures a space after EVERY number,
28             # including the very last one.
29             print(fibonacci(i), end=" ")
30
31  Terminal Test cases
```

1.2.3. Pattern - 1 03.32

Write a Python program to print a pattern of asterisks in the form of a right-angled triangle.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format

- The output should display the pattern of asterisks (*), with each row containing an increasing number of asterisks.

Note: Refer to the displayed test cases for the sample pattern.

Sample Test Cases +

rightangl... SUBMIT

Explorer Debugger

```
1 #Type Content here...
2 def print_star_pattern():
3     try:
4         # Input: Number of rows
5         n = int(input())
6
7         if n <= 0:
8             return
9
10        # Outer loop for the number of rows
11        for i in range(1, n+1):
12            # Inner loop to print stars in each row
13            for j in range(i):
14                # Print star followed by a space
15                print("*", end=" ")
16
17            # Move to the next line after the row is finished
18            print()
19
20        except ValueError:
21            pass
22
23        if __name__ == "__main__":
24            print_star_pattern()
```

Terminal Test cases

PREV RESET SUBMIT NEXT

1.2.4. Pattern - 2 04:34

Write a Python program to print a right-angled triangle pattern of numbers.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format:

- The output should display the pattern of numbers separated by space, with each row containing increasing numbers starting from 1 up to the row number.

Note:

- Refer to the displayed test cases for the sample pattern.

Sample Test Cases +

numberP... SUBMIT

Explorer

```
1 # Type Content here...
2 def print_number_triangle():
3     try:
4         # Input: Number of rows
5         n = int(input())
6
7         if n <= 0:
8             return
9
10        # Outer loop for each row
11        for i in range(1, n + 1):
12            # Inner loop for numbers in each row
13            for j in range(1, i + 1):
14                # Print the number followed by a space
15                # end=" " prevents the default newline behavior
16                print(j, end=" ")
17            # Print a newline after each row is complete
18            print()
19
20
21 except ValueError:
22     print("Invalid input. Please enter an integer.")
23
24 if __name__ == "__main__":
25     print_number_triangle()
26
27
```

Terminal Test cases

Debugger